

Practice Questions Solutions

1.7. Practice Questions

(1-1) Write a program that outputs a string of your choice.

```
print("Hello World!")

Hello World!
```

Example: When $x = \text{"Hello"}$ and $y = \text{"Mr. Matsuo"}$, print $x+y$.

```
x = "Hello"
y = "Mr. Matsuo"
print(x + y)

HelloMr. Matsuo
```

(1-2) Please write one or more programs performing any operations of your choice using arithmetic operators (+, -, *, /, %, **, //).

```
a = 1
b = 2

# addition
print("Addition: ", a + b)

# subtraction
print("Subtraction: ", a - b)

# multiplication
print("Multiplication: ", a * b)

# division
print("Division: ", a / b)

# integer division
print("Integer division: ", a // b)

# remainder
print("Remainder: ", a % b)

# Exponential
print("Exponential: ", a ** 2)
```

```
Addition: 3
Subtraction: -1
Multiplication: 2
Division: 0.5
Integer division: 0
Remainder: 1
Exponential: 1
```

(1-3) Let's assume that an integer is stored in the variable x . Create a program that will display (print) 0 if x is even, and 1 if x is odd.

```
x = 1 # Input example
```

```
print(x % 2)

1
```

(1-4) Create a program to calculate BMI from height and weight. $BMI = \text{weight(kg)} / (\text{height(m)})^2$

```
# Input example
weight = 50 # kg
height = 165 # cm
```

```
bmi = weight / ((height / 100) ** 2)
print(bmi)
```

```
18.36547291092746
```

2.5 Practice Questions

(2-1) Write a list that includes all prime numbers less than 10.

```
primes_list = [2, 3, 5, 7]
print(primes_list)
```

```
[2, 3, 5, 7]
```

(2-2) Write a tuple that includes all prime numbers less than 10.

```
primes_tuple = (2, 3, 5, 7)
print(primes_tuple)
```

```
(2, 3, 5, 7)
```

(2-3) Write a dictionary whose key is the string 'name' and whose value is a string containing your own name.

```
name = {'name': "Taro"}
print(name)
```

```
{'name': 'Taro'}
```

3.3. Practice Questions

(3-1) Write a program using the `if` statement.

```
a = 1
if a % 2 == 0:
    print("Even")
else:
    print("Odd")
```

```
Odd
```

(3-2) Write a program of your choice using a `for` loop.

```
sample_list = [1, 2, 3, 4, 5]

for i in sample_list:
    print(i % 2)
```

```
1
0
1
0
1
```

(3-3) Let's assume that a variable `x` contains a list of integers with a length of 1 or more. Create a program that returns the largest element in `x`. Achieve this without using functions like `max()`, and instead use loops (repetition with `for` statement).

```
x = [1, 2, 3] # Input example
```

```
largest = x[0]
for i in x:
    if i > largest:
        largest = i
print(largest)
```

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4.2. Practice Questions

(4-1) Define a function `f` that takes a string `x` with a length of 1 or more as an argument and returns the last character of `x`.

```
x = 'python'
```

```
def last_char(x):  
    return x[-1]
```

```
last_char(x)
```

```
'n'
```

(4-2) Define a function `f` that takes a dictionary as an argument `x` and returns the number of keys stored in `x`.

```
x = {"a": 1, "b": 2, "c": 3, 'd':4, 'e':5}
```

```
def num_keys(x):  
    return len(x.keys())
```

```
num_keys(x)
```

```
5
```

(4-3) Define a function `f` that takes a list as an argument `x` and returns a list of elements of `x` in reverse order. However, achieve this without using functions like `reversed`, but using loops or list slicing.

```
x = [3, 9, 7, 1, 0]
```

```
def f(x):  
    y = []  
    for i in range(len(x)):  
        y.append(x[-i-1])  
    return y
```

```
f(x)
```

```
[0, 1, 7, 9, 3]
```

(4-4) The following function was intended to display "I am X years old." However, as it is, it results in an error. Please modify the function below. The argument for the function, `age`, is expected to be an integer. Assign the string you want to display to `msg`, store it in a dictionary. Then, print the value from the dictionary by specifying its key.

```
def print_my_age(age):  
    msg = 'My age is ' + str(age) + ' years old.'  
    msg_dict = {'msg': msg}  
    print(msg_dict['msg'])
```

```
print_my_age(18)
```

```
My age is 18 years old.
```

Start coding or [generate](#) with AI.

